

Lab 3 - Routing Protocol

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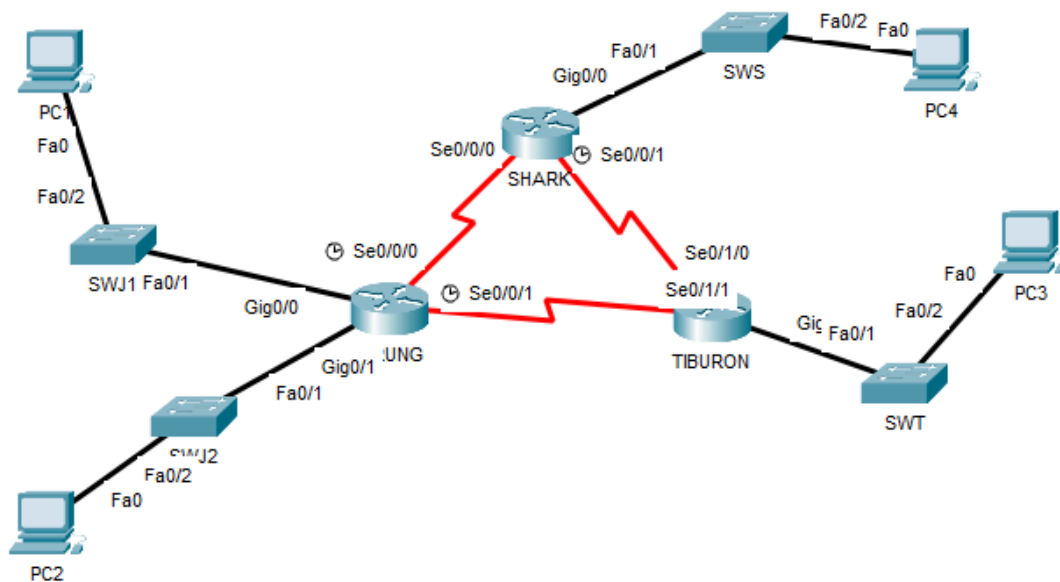
MATRIC NUMBER : A23CS0258

SECTION: 12

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	-
2		Se0/0/1	172.18.110.197	255.255.255.252	-
3		G0/0	172.18.110.1	255.255.255.224	-
4		G0/1	172.18.110.33	255.255.255.224	-
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	-
6		Se0/1/0	172.18.110.202	255.255.255.252	-
7		G0/0	172.18.110.129	255.255.255.192	-
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	-
9		Se0/0/1	172.18.110.201	255.255.255.252	-
10		G0/0	172.18.110.65	255.255.255.192	-
11	PC1	-	172.18.110.30	255.255.255.224	172.18.110.1
12	PC2	-	172.18.110.62	255.255.255.224	172.18.110.33
13	PC3	-	172.18.110.190	255.255.255.192	172.18.110.129
14	PC4	-	172.18.110.126	255.255.255.192	172.18.110.65

LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
- Using the IP addresses calculated, configure the devices with the appropriate information.

Jerung LAN1

- 20 usable host ($2^5 = 32$ IP, 30 usable)
- subnet mask : /27 (255.255.255.224)
- subnet range : 172.18.110.0 - 172.18.110.31 → PC1
- usable IP : 172.18.110.1 - 172.18.110.30
- IP G/O : 172.18.110.1 → gateway PC1

Jerung - Shark

- 2 usable host ($2^2 = 4$ IP, 2 usable)
- subnet mask : /30 (255.255.255.252)
- subnet range : 172.18.110.192 - 172.18.110.195
- usable IP : 172.18.110.193 - 172.18.110.194
jerung shark

Jerung LAN2

- 20 usable host ($2^5 = 32$ IP, 30 usable)
- subnet mask : /27 (255.255.255.224)
- subnet range : 172.18.110.32 - 172.18.110.63 → PC2
- usable IP : 172.18.110.33 - 172.18.110.62
- IP G/O : 172.18.110.33 → gateway PC2

Jerung - Tiburon

- 2 usable host ($2^2 = 4$ IP, 2 usable)
- subnet mask : /30 (255.255.255.252)
- subnet range : 172.18.110.196 - 172.18.110.199
- usable IP : 172.18.110.197 - 172.18.110.198
jerung tiburon

Tiburon LAN

- 50 usable host ($2^6 = 64$ IP, 62 usable)
- subnet mask : /26 (255.255.255.192)
- subnet range : 172.18.110.128 - 172.18.110.191 → PC3
- usable IP : 172.18.110.129 - 172.18.110.190
- IP G/O : 172.18.110.129 → gateway PC3

Shark - Tiburon

- 2 usable host ($2^2 = 4$ IP, 2 usable)
- subnet mask : /30 (255.255.255.252)
- subnet range : 172.18.110.200 - 172.18.110.203
- usable IP : 172.18.110.201 - 172.18.110.202
shark tiburon

Shark LAN

- 60 usable host ($2^6 = 64$ IP, 62 usable)
- subnet mask : /26 (255.255.255.192)
- subnet range : 172.18.110.64 - 172.18.110.127 → PC4
- usable IP : 172.18.110.65 - 172.18.110.126
- IP G/O : 172.18.110.65 → gateway PC4

Task 2 – Routing Table

- Paste the current routing table of each router here. There are 2 ways to do this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - CLI

- i. Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

Figure A

```
SHARK>enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.65/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

SHARK#
```

Figure a.1 Shark

```
TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

      172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.128/26 is directly connected, GigabitEthernet0/0
L       172.18.110.129/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

TIBURON#
```

Figure a.2 Tiburon

```

JERUNG>enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
       C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
       L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
       C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
       L       172.18.110.33/32 is directly connected, GigabitEthernet0/1
       C       172.18.110.192/30 is directly connected, Serial0/0/0
       L       172.18.110.193/32 is directly connected, Serial0/0/0
       C       172.18.110.196/30 is directly connected, Serial0/0/1
       L       172.18.110.197/32 is directly connected, Serial0/0/1

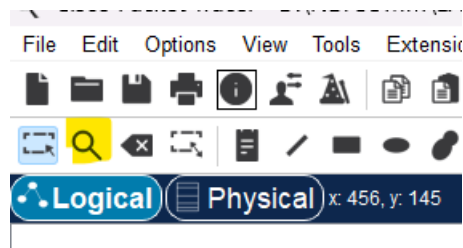
JERUNG#

```

Figure a.3 Jerung

b. PT tool

- i. Click on the magnifying glass (shown in Figure B), then click on a router, then choose the Routing Table. A sample of the result is shown in Figure C.

*Figure B*

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0

Figure C

Routing Table for SHARK				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

Figure b.1 Shark

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Figure b.2 Tiburon

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0

Figure b.3 Jerung

2. Try to ping PC2 from PC1, paste the results here.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.62

Pinging 172.18.110.62 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127
Reply from 172.18.110.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

3. Try to ping PC4 from PC1, paste the results here.

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

4. Explain the reason(s) behind the results.

PC1 cannot ping PC4 because PC4 and PC1 are both connected to different routers which are not interconnected. The packets shared between the subnets are not forwarded as there is no way into the subnet.

5. What needs to be done to ensure all PCs can ping each other successfully?

By making sure that all routers are connected by RIP.

Task 3 – Routing Configuration

1. Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.128/27	GigabitEthernet0/1	---	0/0
L	172.18.110.129/32	GigabitEthernet0/1	---	0/0
C	172.18.110.160/27	GigabitEthernet0/0	---	0/0
L	172.18.110.164/28	GigabitEthernet0/0	---	0/0

Figure D

Routing Table for TIBURON					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0	
C	172.18.110.200/30	Serial0/1/0	---	0/0	
L	172.18.110.202/32	Serial0/1/0	---	0/0	

Routing Table for JERUNG					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.193/32	Serial0/0/0	---	0/0	
C	172.18.110.196/30	Serial0/0/1	---	0/0	
L	172.18.110.197/32	Serial0/0/1	---	0/0	

Figure 1. Tiburon and Jerung before RIP

2. In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

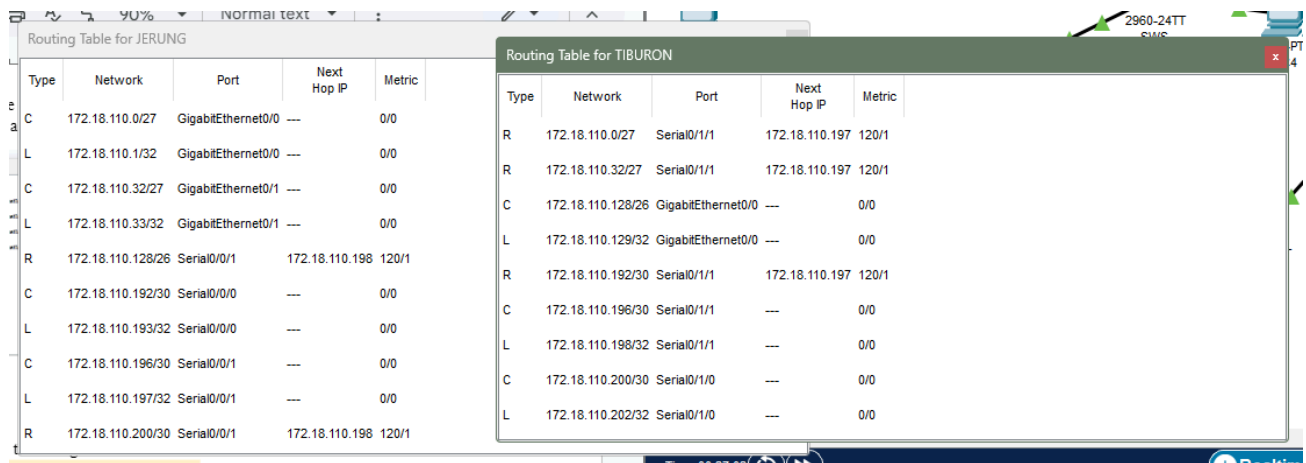
- a. What can you say about the addresses used in the 'network' instructions in Figure E?

The network instructions apply subnet addresses to determine which networks that RIP should advertise and include in its routing updates. The 'no auto-summary' command disables classful analysis of network boundaries, in order to maintain more specific routing information.

3. Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```
TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200
```

4. As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.



The image shows two side-by-side screenshots of router routing tables. The left screenshot is titled 'Routing Table for JERUNG' and the right is titled 'Routing Table for TIBURON'. Both tables have columns for Type, Network, Port, Next Hop IP, and Metric.

Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.33/32	GigabitEthernet0/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.198	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.18.110.198	120/1

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.128/26	GigabitEthernet0/0	---	0/0
L	172.18.110.129/32	GigabitEthernet0/0	---	0/0
R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.196/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

Figure 2. Tiburon and Jerung after RIP

- a. What are the changes seen in TIBURON?

There's 3 R types added into Tiburon.

- b. What are the Networks with type R in TIBURON and JERUNG?

In Tiburon; 172.18.110.0/27, 172.18.110.32/27, and 172.18.110.192/30.

In Jerung; 172.18.110.128/26 and 172.18.110.200/30.

- c. Ping PC3 from PC1. Was it successful?

```

C:\>ping 172.18.110.190

Pinging 172.18.110.190 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=1ms TTL=126
Reply from 172.18.110.190: bytes=32 time=2ms TTL=126

Ping statistics for 172.18.110.190:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

```

Yes, it is successful.

- d. Ping PC4 from PC1. Was it successful?

```

C:\>ping 172.18.110.126

Pinging 172.18.110.126 with 32 bytes of data:

Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Reply from 172.18.110.1: Destination host unreachable.
Request timed out.

Ping statistics for 172.18.110.126:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

No, it failed.

- e. Explain the reasons for your answer.

The routers of PC1 and PC4, which are SHARK and JERUNG respectively, are not connected. Therefore packets to be sent are dropped along the way.

- f. Continue with configuration of RIP in SHARK. Paste your configurations here.

```

SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.64
SHARK(config-router)#network 172.18.110.128
SHARK(config-router)#network 172.18.110.192
SHARK(config-router)#no auto-summary
SHARK(config-router)#

```

- g. Open router SHARK's routing table and paste here.

Type	Network	Port	Next Hop IP	Metric
C	171.18.110.192/30	Serial0/0/0	---	0/0
L	171.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.0/27	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.32/27	Serial0/0/1	172.18.110.202	120/2
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/1
R	172.18.110.192/30	Serial0/0/1	172.18.110.202	120/2
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- h. Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	<pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=23ms TTL=125 Reply from 172.18.110.30: bytes=32 time=3ms TTL=125 Reply from 172.18.110.30: bytes=32 time=2ms TTL=125 Reply from 172.18.110.30: bytes=32 time=2ms TTL=125 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 23ms, Average = 7ms C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=16ms TTL=125 Reply from 172.18.110.30: bytes=32 time=27ms TTL=125 Reply from 172.18.110.30: bytes=32 time=14ms TTL=125 Reply from 172.18.110.30: bytes=32 time=20ms TTL=125 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 14ms, Maximum = 27ms, Average = 19ms </pre>

2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=2ms TTL=125 Reply from 172.18.110.62: bytes=32 time=23ms TTL=125 Reply from 172.18.110.62: bytes=32 time=23ms TTL=125 Reply from 172.18.110.62: bytes=32 time=20ms TTL=125 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 23ms, Average = 17ms C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=3ms TTL=125 Reply from 172.18.110.62: bytes=32 time=2ms TTL=125 Reply from 172.18.110.62: bytes=32 time=2ms TTL=125 Reply from 172.18.110.62: bytes=32 time=13ms TTL=125 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 13ms, Average = 5ms </pre>
3	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=14ms TTL=128 Reply from 172.18.110.126: bytes=32 time=4ms TTL=128 Reply from 172.18.110.126: bytes=32 time=2ms TTL=128 Reply from 172.18.110.126: bytes=32 time=3ms TTL=128 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 14ms, Average = 5ms C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=15ms TTL=128 Reply from 172.18.110.126: bytes=32 time=4ms TTL=128 Reply from 172.18.110.126: bytes=32 time=16ms TTL=128 Reply from 172.18.110.126: bytes=32 time=14ms TTL=128 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 4ms, Maximum = 16ms, Average = 12ms </pre>

Task 4 – Routing Update

1. Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - a. This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0/24
 - b. As this change happens, PC3 must also have a different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.2
Subnet Mask	255.255.255.0
Gateway Address	192.168.1.1

- c. After this change, can PC4 and PC1 ping PC3?
No.

- d. Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	171.18.110.192/30	Serial0/0/0	---	0/0	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
L	171.18.110.194/32	Serial0/0/0	---	0/0	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.0/27	Serial0/0/1	172.18.110.202	120/2	R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
R	172.18.110.32/27	Serial0/0/1	172.18.110.202	120/2	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.128/26	Serial0/0/1	172.18.110.202	120/16	C	172.18.110.200/30	Serial0/1/0	---	0/0
R	172.18.110.192/30	Serial0/0/1	172.18.110.202	120/2	L	172.18.110.202/32	Serial0/1/0	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	L	192.168.1.1/32	GigabitEthernet0/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0					

- e. Referring to the routing table, explain your findings.

The IP Address was changed causing connection to be lost between networks.

- f. What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

I have to update router configuration and add 192.168.1.0/24 to the RIP.

- g. Show your configurations in TIBURON to ensure end-to-end connectivity.

```
TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#no auto-summary
TIBURON(config-router)#
```

- h. To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre>Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.30 Pinging 172.18.110.30 with 32 bytes of data: Reply from 172.18.110.30: bytes=32 time=2ms TTL=126 Reply from 172.18.110.30: bytes=32 time=12ms TTL=126 Reply from 172.18.110.30: bytes=32 time=17ms TTL=126 Reply from 172.18.110.30: bytes=32 time=2ms TTL=126 Ping statistics for 172.18.110.30: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 2ms, Maximum = 17ms, Average = 8ms</pre>

2	<pre> C:\>ping 172.18.110.62 Pinging 172.18.110.62 with 32 bytes of data: Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=1ms TTL=126 Reply from 172.18.110.62: bytes=32 time=21ms TTL=126 Reply from 172.18.110.62: bytes=32 time=10ms TTL=126 Ping statistics for 172.18.110.62: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 21ms, Average = 8ms </pre>
4	<pre> C:\>ping 172.18.110.126 Pinging 172.18.110.126 with 32 bytes of data: Reply from 172.18.110.126: bytes=32 time=2ms TTL=126 Reply from 172.18.110.126: bytes=32 time=11ms TTL=126 Reply from 172.18.110.126: bytes=32 time=11ms TTL=126 Reply from 172.18.110.126: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.126: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 11ms, Average = 6ms </pre>

REFLECTION

What have you learned in this task?

This task taught me about hands-on practice of the IP address and subnetting, namely calculating subnet addresses and providing usable IP addresses to devices. The importance of configuring routers using both CLI and Packet Tracer lies in knowing how to parse forwarding tables and, lastly, how routing protocols like RIP make possible the end-to-end delivery of packets. The job also highlighted troubleshooting through scenarios where the wrong IP addresses or disconnections of routers result in the loss of packets, requiring changes on router settings and routing protocols, emphasizing how the subnets configurations affect network communication. Overall, the practically learned material was effective in improving my grasp of the data-plane capabilities of the network layer, IP routing, and the main network troubleshooting techniques.